OPERATING SYSTEMS REVISION NOTES - EXAM CHEATSHEET

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1. PROCESS MANAGEMENT AND SCHEDULING ALGORITHMS

- FCFS (First Come First Serve): Non-preemptive, suffers from convoy effect.
- SJF (Shortest Job First): Optimal average waiting time, but requires knowing burst time.
- SRTF (Shortest Remaining Time First): Preemptive version of SJF.
- Round Robin (RR): Time quantum based, preemptive, fair responsiveness for interactive systems.

2. MEMORY MANAGEMENT AND VIRTUAL MEMORY

- Logical vs Physical Address Space: MMU translates virtual page numbers to physical frames.
- TLB (Translation Lookaside Buffer): Associative hardware cache for fast virtual-to-physical address mapping.
- Demand Paging: Pages are loaded into RAM only when accessed. Page fault triggers OS interrupt.
- Page Replacement Policies: LRU (Least Recently Used), FIFO, Optimal (Beladys anomaly proof).

3. SYNCHRONIZATION AND DEADLOCK PREVENTION

- Critical Section Problem: Requirements are Mutual Exclusion, Progress, and Bounded Waiting.
- Semaphores: Counting and binary mutex semaphores using atomic wait() and signal() primitives.
- Deadlock Coffman Conditions: Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.
- Prevention: Break circular wait by imposing strict global ordering on resource allocation.